

PH3202 (Intermediate EM)

Instructor: Golam M. Hossain

Class Test 3

Question 1 (3 marks)

Show that the Maxwell's tensor $F_{\mu\nu}$ is invariant under gauge transformation.

Question 2 (7 marks)

An observer in O frame observes charge density to be ρ_0 and no current density, i.e., $\vec{J} = 0$. Using Lorentz transformation, find the charge and current density in O' frame, which is moving at a speed v in +ve x direction wrt O.